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Research on Optimization of Urban Logistics Network Based on Low Altitude Economy: Path Planning and Traffic Impact Analysis of Collaborative Distribution Mode Between Drones and Ground Vehicles

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Abstract

As urban development accelerates, the limitations of traditional logistics are becoming increasingly apparent. The low altitude economy has shown its importance under the current circumstances. Most of the research on this new type of transportation focuses only on single distribution mode, which lacks the analysis of collaborative distribution. In this study, we build a network which consists a single distribution center and multiple customer points with fixed demand. We aim to explore the most efficient way within a specific distribution area. A total of three kinds of delivering groups are set up: pure drone distribution (Unmanned Aerial Vehicle Routing Problem); pure ground vehicle distribution (Capacitated Vehicle Routing Problem); collaborative drone and vehicle distribution (Vehicle Routing Problem with Drones). We also propose a model to maximize the overall distribution efficiency under several assumptions. In order to get the highest distribution efficiency, we set efficiency as a weighted composite of time and cost. This model combines the constraints and indicators such as capacity and travel distance. Moreover, by using a hybrid algorithm with Mixed Integer Linear Programming, our algorithm design utilizes an innovative solution to solve the non-deterministic polynomial hard problem. To verify the viability of the theoretical model and provide quantifiable comparisons to support our conclusions, we also implement a case study in Shenzhen, China. It shows that the collaborative distribution model outweighs the vehicle-only model and the unmanned aerial vehicle-only model in terms of total trips, delivery time, and total cost. This has proven to be the most efficient way to get the most out of distribution.

Keywords: low altitude economy; Drone – Ground Vehicle Collaborative Delivery; mixed-mode logistics; Last-mile Logistics; Mixed-integer linear programming

Contents

Abstract 3

1. Introduction 5

2. Literature Review 9

3. Mathematical Model 12

- **3.1 Description of the problem** 12
- **3.2 Model assumption** 15
- **3.3 Description of notation** 17
- **3.4 Mathematical model formulas** 19

4. Algorithm Design 21

- **4.1 Overall design of the algorithm** 21
- **4.2 Mixed Integer Linear Programming (MILP)** 23
- **4.3 Gurobi solver applications** 24
- **4.4 Traveler's Nearest Neighbor Heuristic Algorithm (TSP)** 25
- **4.5 Genetic Algorithm** 27
- **4.6 Warm Start** 30

5. Case study in Shenzhen 31

- **5.1 Parameter settings** 31
- **5.2 Results** 33
- **5.3 Sensitivity test analysis** 37

6. Discussion 39

7. Recommendation 42

8. Conclusions 44

Acknowledgements 46

References 47

1. Introduction

With the rapid advancement of urbanization, cities are taking on increasingly complex and diverse functions. However, conventional transportation systems are struggling to keep pace with these evolving demands. In particular, traditional logistics networks often lack the flexibility and efficiency required for time-sensitive or high-risk tasks—such as the rapid delivery of medical supplies, emergency response operations, or precision agricultural services. These limitations highlight an urgent need for innovative delivery solutions, which are capable of addressing the shortcomings of existing systems.

In response to these growing logistical challenges, the emergence of the low-altitude economy presents a promising alternative. Generally, The low-altitude economy encompasses a growing set of economic activities that utilize the low-altitude airspace——within 1,000 meters of the vertical distance from the ground. The number can be extended to 3,000 meters in different regions with different functions. Aerial vehicles play a central role in enabling this economic model. Among them, drones stand out for their advantages in operational flexibility, high delivery efficiency, and reduced reliance on land-based infrastructure. Driven by advances in unmanned aerial vehicle (UAV) technology and autonomous ground transportation, this new economic paradigm is reshaping the way goods and services move within cities. Drone-based logistics represents a cost-effective solution to enhance delivery efficiency and adaptability, particularly in areas where traditional transportation faces spatial or temporal constraints. For rural areas where part of the transportation infrastructure is quite inadequate, drones facilitate logistics and distribution in hard-to-reach areas, consolidating the outcomes of poverty alleviation initiatives and supporting the upgrading of local industries. For urban areas, where the e-commerce and takeaway industry is booming, it can provide more convenient services for the consumers.

Compared with traditional transportation network in the cities, the most distinctive advantage of low-altitude logistics is its utilization of the vertical, or third, spatial dimension. A logistics model that relies on three-dimensional spatial development provides support for low-altitude logistics to accommodate higher transportation throughput by alleviating ground-level congestion and optimizing spatial resource allocation. Increasingly supported by national and municipal policy frameworks, the low-altitude economy, often solve the problem in term of the "sixth dimension" of urban space, integrates underutilized airspace into logistics and planning strategies. It introduces an additional spatial dimension to urban logistics planning, utilizing previously underutilized airspace. In China, both national and municipal governments have put forward clear plans to develop the low-altitude economy, aiming to promote the use of general aviation and UAVs in logistics, emergency rescue, and other fields, including construction of a low-altitude flight service guarantee system (State Council of China, 2021; Shenzhen Municipal People’s Congress, 2024). The General Office of the State Council, PCR and the Central Military Commission (2010) issued the Opinions on Deepening the Reform of China's Low Altitude Airspace Management. It is the first time to clarify the overall objectives, phases and main tasks of deepening the reform of low altitude airspace management. In 2016, the General Office of the State Council issued the Guiding Opinions on Promoting the Development of the General Aviation Industry, which positioned the general aviation industry as a strategic and emerging industrial system. The Government Work Report of the National People's Congress of the State Council (2024) included "low-altitude economy" for the first time.

Against this backdrop, drone and ground unmanned vehicle technology has progressed quickly, transitioning from conceptual frameworks to practical, commercialized applications. Taking companies such as Meituan and Jingdong as examples, they have taken the lead in trying to use drones and unmanned vehicles to collaborate on delivery (Fig. 1). This not only shortens the delivery

time, but also reduces the cost of logistics labor. Meanwhile, academic researchers are also exploring this topic. The current research on low-altitude logistics mainly focuses on delivery path optimization or independent distribution by drones instead of optimal Utilization of Drones and Other Means of Transportation. Existing studies have explored road network accessibility based on spatial syntax theory, and multi-objective scheduling algorithms for the linkage of drones and vehicles (Otto et al., 2018). However, there are still significant gaps in the existing research: most of the studies are limited to the optimization of a single distribution tool, and there is a lack of comprehensive and quantitative analysis on the efficiency comparison between the collaborative mode and the traditional vehicle-only or UAV-only mode.

Fig. 1. Urban air logistics application scenarios

(Retrieved from: https://mp.weixin.qq.com/s/z3zXobHwI_8RbGSK3Y9Fcg)

To fully leverage the complementary strengths of ground vehicles and UAVs, this study has designed an integrated delivery framework that combines the two distribution modes. This framework coordinates the advantages of ground

vehicles with large loads and long range, and the characteristics of UAVs with fast response and flexibility. By coordinating these capabilities, the suggested model intends to improve overall delivery efficiency. To narrow the gap in the existing literature, our approach commences deliveries from a distribution center, where both UAVs and ground vehicles are deployed concurrently to serve several fixed-demand nodes before returning to the origin location. This parallel deployment strategy lays the groundwork for a revolutionary collaborative distribution mechanism.

This collaborative distribution mode originates from the classical Vehicle Path Problem (VRP). This is about planning an optimal distribution route to reduce costs while satisfying constraints such as load and distance (Dantzig & Ramser, 1959). This model plays an important role in our study. The VRP model has developed several variants so that it could adapt to different realistic scenario constraints. The following are a few types of classic variants: 1. CVRP (Capacitated Vehicle Routing Problem): Considers vehicle load constraints and requires that the sum of customer demands on each path does not exceed the vehicle capacity. 2. VRPTW (Vehicle Routing Problem with Time Windows): Customer service time window constraints, and the vehicle must arrive at the customer point within a specified time frame. 3. VRPPD (Vehicle Routing Problem with Pickup and Delivery): Pickup and delivery sequence & coordinate path. 4. VRPD (Vehicle Routing Problem with Drones): Optimizing paths in conjunction with ground-based vehicles. 5. SDVRP (Split Delivery Vehicle Routing Problem): Allows individual customer demand to be delivered in multiple deliveries by multiple vehicles.

In this thesis, we set three delivery groups: a pure drone group (Unmanned Aerial Vehicle Routing Problem), a pure ground vehicle group (Capacitated Vehicle

Routing Problem), and a mixed group (Vehicle Routing Problem). In the third group, the UAVs and vehicles are operated independently but collaborate with each other during the same delivery cycle. The research goal is to maximize delivery efficiency, which is the weighted sum of the total delivery time and transportation cost.

Given the complexity of the problem, a hybrid algorithm is designed to obtain an optimal or near-optimal solution. The algorithm uses mixed integer linear programming (MILP). The algorithm integrates mixed integer linear programming (MILP) with heuristic approaches. Specifically, this study applies a genetic algorithm (GA) in combination with a nearest neighbor heuristic to address the computational challenges associated with large-scale distribution path planning. The effectiveness of the proposed hybrid algorithm is validated through a case study conducted in Shenzhen. The result provides insights into the potential benefits of combining ground vehicles and UAVs for distribution. It also highlights the advantages of the hybrid distribution mode in terms of shorter transportation distances, shorter delivery times, and lower total costs.

In sum, while a few existing studies have explored modes of UAV-assisted vehicle distribution, they lack a comprehensive assessment of the performance of the collaborative distribution mode system and its impact on the prior and future development of the low-altitude economy. To fill the above gaps, this paper proposes an urban logistics and distribution optimization model for UAV-vehicle collaboration with ground vehicles, and designs a hybrid algorithm to solve it.

2. Literature Review

The low-altitude economy is emerging as a new spot in global economic

development. With the rapid development of aviation technology, information technology, new energy technology, and other fields, the performance of low-altitude vehicles has been improving, and the cost has been gradually reduced (Thipphavong et al., 2018) At the same time, society's growing demand for efficient, convenient, and personalized transportation and logistics has also provided a broad market space for low-altitude economy.

Despite the huge development potential of low-altitude economy, it also faces many challenges. Airspace management is one of the key factors restricting its development. At present, the management of low-altitude airspace in most regions of the world is still relatively strict (Dorling et al., 2017). For example, Shenzhen has restricted some flight airspace of UAVs through the Interim Measures for the Management of Civilian Micro and Light Drones (Shenzhen Municipal Government, 2023). In addition, the safety and reliability of low-altitude vehicles are also the problems that need to be addressed. Due to the uncertainty of the low-altitude flight environment, drones are easily affected by weather, air obstacles, and many other factors (Clothier et al., 2015).

In order to promote the low altitude economy, governments have introduced policy measures. For example, the United States Federal Aviation Administration (FAA) launched the Urban Air Mobility (UAM) program, which aims to promote the development of urban air transportation through technological innovation and policy support (FAA, 2023). China has explicitly proposed the development of low-altitude economy in the "National comprehensive three dimensions transportation network planning outline." and has launched pilot low-altitude reforms in Hainan and Jiangxi Province (State Council of China, 2021).

The rise of unmanned aerial vehicles is a trigger in shaping the low-altitude economy (Ren et al., 2021). Integrating drones into existing logistics and supply chain systems has the potential to increase efficiency, reduce costs, and improve

customer satisfaction (Goodchild & Toy, 2018). However, despite some outcomes, drone delivery still has technical constraints. For instance, they have limited payload capacity and battery life. Thus, they may only able to serve enough amount of customers in a single flight (Agatz et al., 2018). In addition, it also requires some adjustments to operational procedures and infrastructure (Murray & Raj, 2020) But this can be expensive and time-consuming (Figliozzi, 2010).

There is a lot of literature on UAV transportation. Small and Verhoef (2007) stated that it is important to assess the competitive advantages of drone delivery. If drones can provide faster delivery times and greater reliability, users may be more willing to choose this mode. This can hugely influence the design of overall logistics network. The joint logistics model of drones and ground vehicles can be divided into four. There are vehicle-guaranteed drone distribution, drone-guaranteed vehicle distribution, parallel distribution, and synchronized distribution mode (Li & Wang). Murray and Chu (2015) proposed a new model called TSP to study the optimization problem of UAV-assisted parcel delivery. The model adopts a hybrid distribution strategy, which means UAVs and ground vehicles can work together to minimize the total distribution time and cost. This provides a theoretical basis and practical guidance for us to examine hybrid distribution modes. Wang et al. (2017) designed a hybrid algorithm for solving the Vehicle Path Problem with Drones (VRP with Drones). The algorithm combines the advantages of exact and heuristic algorithms to find a high quality solution in a reasonable time. Agatz et al. (2018) explores the application of drones in the traveler's problem and considers the navigational constraints of drones, such as the maximum load, maximum range, and so on. Through mathematical modeling and algorithm design, the study provides theoretical support for the optimization of UAV delivery. Beside, in most cases, drones can finish delivery in a shorter time, in other words, they can do more deliveries in the same time. Clearly, this is more environmental friendly (Tamke & Buscher,

2021).

There are also some shortcomings in the literature. Li & Xin (2024) assumed the distance between points in a straight line when distributing distances. And they did not take into account the effects of 3D spatial factors such as buildings and terrain in the city on the flight paths of UAVs. In practice, drones may need to fly in three-dimensional space to avoid obstacles, which may increase the flight distance and time. Ma et al. (2024) extends the experiment based on Solomon's example, but does not adjust the data. For example, there are distribution of forbidding fly zones for drones, the proportion of customer types, and so on. To meet the need for "truck + drone" cooperation, this may affect the generality of the conclusions.

Given the existing literature and challenges, this study aims to propose an integrated mathematical model. Accordingly, we can optimize the collaboration between ground vehicles and drones in distribution operations. By mixing drones and ground trucks, the problems mentioned before can be relatively alleviated. In addition, by introducing Gurobi solving tools to accelerate the convergence of the algorithm, the method proposed in this study is more concise and easier to execute compared to traditional algorithms.

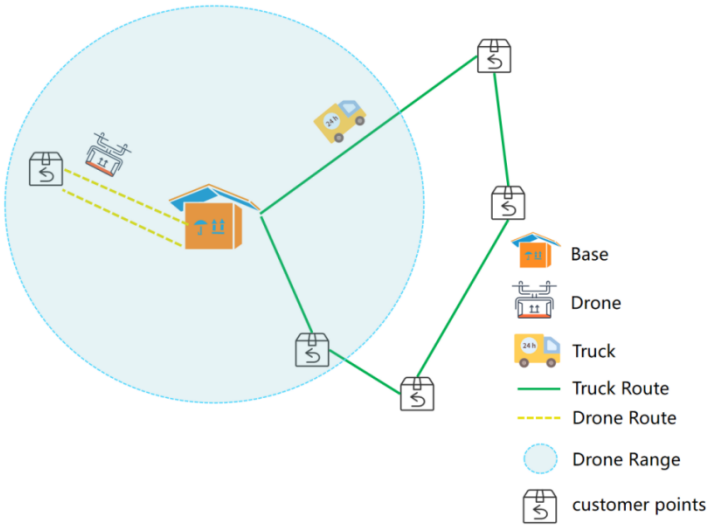
3. Mathematical Model

3.1 Description of the problem

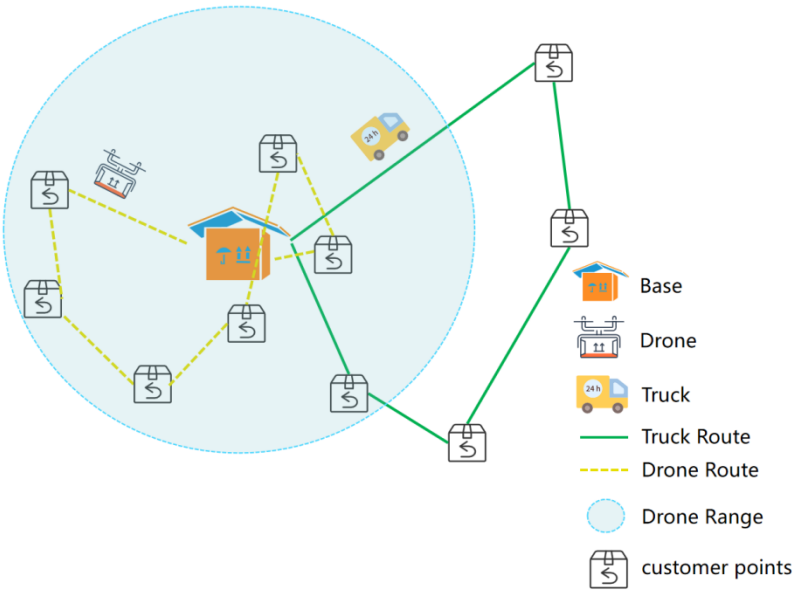
This study investigates the fixed distribution of a distribution center and several customer points within a delivery area, where each customer's demand is predetermined and has a consistent upper limit. Both ground vehicles and drones originate from and return to the distribution center, employing independent delivery methods to fulfill logistics tasks. Three delivery modes are constructed by the model: the pure Unmanned Aerial Vehicle Routing Problem (UAVRP), the

pure Capacitated Vehicle Routing Problem (CVRP), and the Vehicle Routing Problem with Drones (VRPD), which involves the collaboration of drones and ground vehicles In order to avoid job interference, the collaborative delivery group uses a parallel independent delivery method in which drones and ground vehicles independently arrange their itineraries (Narayan et al., 2021). As shown in Fig. 2, being distinguished from single-flight, single-point delivery mode, drones utilize a single-flight, multi-point delivery mode, capable of completing multiple customer deliveries in one trip, provided the total demand of the selected customers does not exceed the drone's maximum payload and the flight path length is within the drone's maximum range (Li & Xin, 2024). To ensure the comparability and fairness of experimental results across the three delivery models, all three delivery modes share the same fixed costs, with fixed customer demand and the delivery tools' capacity exceeding the demand of a single customer location. Also for better parameterized configuration and comparison of resultant simulations, Unit distance costs for drones and ground vehicles are based on industry white papers and studies, respectively, taking values within their ranges. Regarding delivery time, the collaborative group's total delivery time is defined as the maximum of the individual delivery times of ground vehicles (calculated using Manhattan distance) and drones (calculated using Euclidean distance) to ensure that the actual delivery time of the collaborative delivery mode is fully considered; the pure drone group and the pure ground vehicle group respectively take the total delivery time of their respective delivery tools. The objective function of the model is the comprehensive optimization of delivery efficiency, that is, to minimize the weighted sum of total delivery time and total transportation cost. At the same time, in order to ensure the rationality and reality of the solution, the model introduces delivery center constraints, flow balance constraints, MTZ sub-tour elimination constraints, load constraints, and drone endurance constraints. These constraints jointly ensure that each customer point is only served once, each path starts from the distribution center and finally returns, each vehicle serves each customer at most once in a single delivery, and

meets the physical performance limitations of ground vehicles and drones. As shown in Fig. 3, It demonstrates the construction logic of our model. The above settings and constraints together provide a rigorous theoretical basis and practical basis for the performance comparison and optimization strategies of different delivery modes.



(1) Single-flight, single-point delivery mode



(2) Single-flight, multi-point delivery mode

Fig. 2. Two delivery models for drones

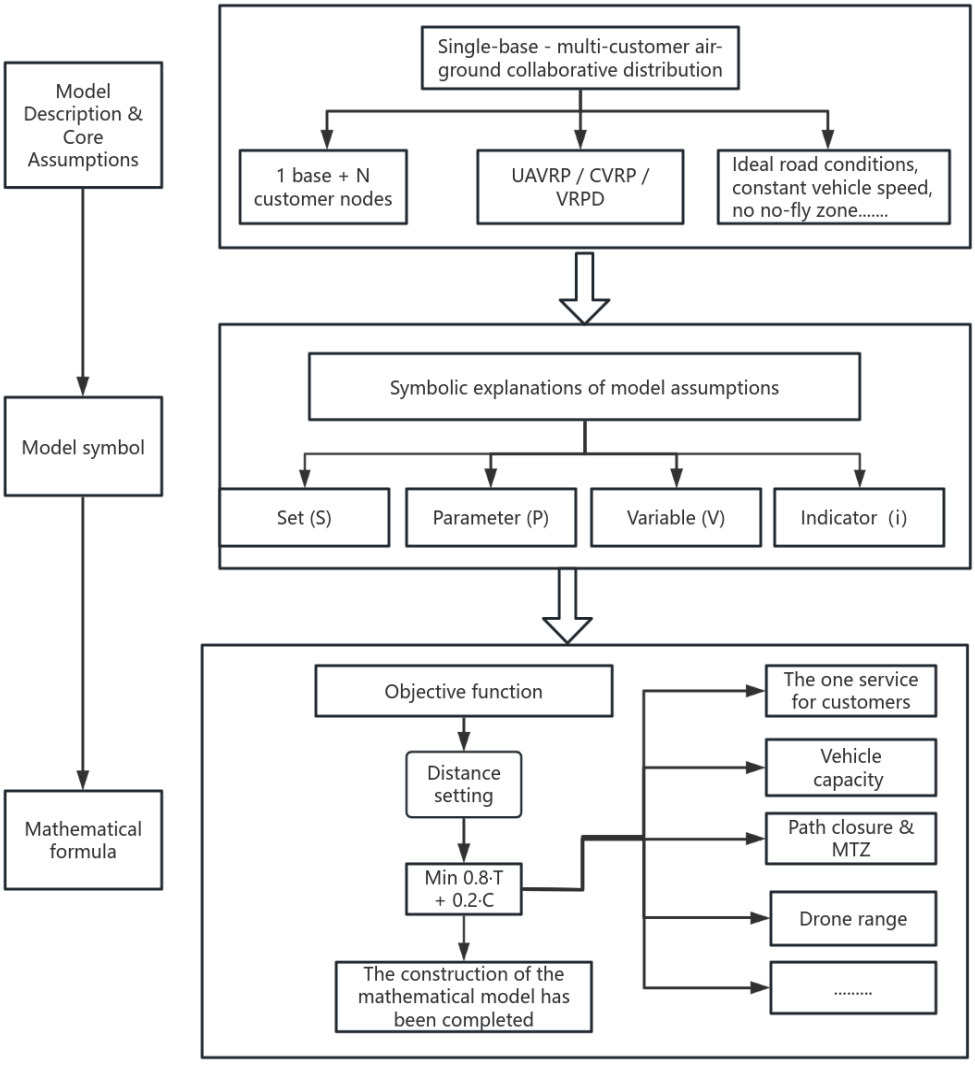


Fig. 3. Model structure

3.2 Model assumption

The following assumptions are made:

(1) All delivery vehicles operate at a constant velocity throughout the delivery process; specifically, both ground vehicles and drones maintain a consistent speed, under the situation that drones operating at a significantly higher speed than ground vehicles. It is assumed that the transport load does not affect the travel or flight speed.

(2) For ground vehicles, only weight limits are taken into account so, every truck has a specified maximum weight capacity; fuel range and maximum trip distance

are not taken into account. For drones, both weight constraints and maximum range constraints are considered.

(3) The delivery process is assumed to occur in an ideal environment and does not account for external factors such as weather conditions, traffic congestion, or changes in road conditions that may occur during real-world deliveries.

(4) The collaborative delivery mode in this paper uses a parallel delivery strategy in which ground vehicles and drones depart from the distribution center inside the same delivery cycle independently. There is no cross-operation or collaborative delivery between the two, and ground vehicles do not carry drones.

(5) Indirect costs, such as wear and tear, maintenance, repairs, parts replacement, and site maintenance, incurred by drones and ground vehicles during operation, are included in their respective per-distance operating costs and fixed costs and are not listed separately.

(6) Each delivery tool consumes a fixed delivery time upon the completion of each delivery task. The fixed delivery time for ground vehicles includes operations such as loading/unloading goods and handover, while the fixed delivery time for UAVs includes loading goods and delivery/unloading operations, but excludes additional waiting or service time at customer points.

(7) All three modes in this study (pure UAV group, pure ground vehicle group, and collaborative group) adopt uniform unit distance transportation costs and fixed cost parameters. The number of ground vehicles and UAVs required for each group is determined based on this preset cost standard.

(8) The distance calculation method between delivery nodes is strictly differentiated as follows: the delivery routes of ground vehicles are calculated using Manhattan distance to simulate the two-dimensional driving paths of urban road networks; the delivery routes of UAVs are calculated using Euclidean distance to reflect the straight-line flight characteristics of UAVs.

(9) The delivery goods demand for each customer points are fixed and consistent, with no significant differences in demand among individual customers; only a clear maximum load demand for a single customer is set.

(10) No restricted access areas or no-fly zones are considered within the delivery area, meaning all areas are assumed to be accessible and allow entry and flight.

(11) Each customer's goods delivery task must be independently completed by a single UAV or a single ground vehicle, and is not allowed to be completed by multiple delivery tools in multiple trips or through collaboration.

(12) Without setting time window constraints for delivery, there are no specific restrictions on the delivery arrival time for customer points, and the arrival of the delivery vehicle at the customer point at any time is considered valid service.

(13) The model assumes that the conversion factor between total delivery time and transportation cost has been normalized, fully considering the time and economic costs of operations, receiving, and maintenance involved in the actual delivery scenario to ensure the consistency of the dimensions of each indicator in the objective function.

3.3 Description of notation

[Set]

$N = \{0,1,2,\dots,n\}$

Node 0 denotes the distribution center, while nodes i=1,2,...,n represent the customers.

$V = \{1,2,\dots,NUM_TRUCKS\}$

The set of ground vehicles.

$U = \{1,2,\dots,NUM_DRONES\}$

The set of drones

[Parameters]

NUM_NODE —— Number of customers (excluding distribution centers)

NUM_TRUCKS —— Number of ground vehicles.

NUM_DRONES —— Number of drones

RANGE_COORDINATE —— Coordinate range

RANDOM_SEED —— Random seed

BIG_M —— Big-M constant (for MTZ constraints)

TRUCK_COST —— Ground vehicle unit distance cost

DRONE_COST —— Drone unit distance cost.

FIXED_COST_TRUCK —— Fixed cost of ground vehicles.

FIXED_COST_DRONE —— Drone fixed costs

TRUCK_CAPACITY —— Maximum payload capacity of ground vehicles.

DRONE_CAPACITY —— Maximum drone payload.

UPPER_WEIGHT_ITEM —— Fixed customer demand.

DRONE_RANGE —— Maximum drone flight range.

TRUCK_SPEED —— Ground vehicle velocity.

DRONE_SPEED —— Drone flight speed

FIXED_TIME_TRUCK —— Fixed delivery times for ground vehicles

FIXED_TIME_DRONE —— Fixed delivery times for drones

TIME_WEIGHT —— Time cost weighting

COST_WEIGHT —— Cost weights

[Distance metrics]

M_{ij} —— Manhattan distance between node i and node j

D_{ij} —— Euclidean distance between node i and node j

[Variable]

$x_{k,i,j}$ —— Binary decision variable whether or not vehicle k moves from node i to node j

$u_{k,i}$ —— Auxiliary variable for ground vehicle k at node i (for MTZ constraints)

$y_{d,i,j}$ —— Binary decision variable whether or not vehicle d moves from node i to node j

$v_{d,i}$ —— Auxiliary variable for drone d at node i (for MTZ constraints)

[Evaluation indicators]

T_{truck}: Total ground vehicle distribution time

T_{drone}: Total drone delivery time

$T_{overall} = \max\{T_{truck}, T_{drone}\}$: Total Delivery Time

C_{truck}: Total ground vehicle transportation costs

C_{drone}: Total drone transportation costs

C_{total}: Total transportation costs

Z: Comprehensive distribution efficiency indicators

3.4 Mathematical model formulas

$$minZ = TIME_WEIGHT \times T_{overall} + COST_WEIGHT \times (C_{truck} + C_{drone}) \quad (1)$$

$$T_{overall} = max\{T_{truck}, T_{drone}\} \quad (2)$$

$$T_{truck} = \frac{60}{TRUCK_SPEED} \sum_{k \in V} \sum_{\substack{i,j \in N \\ i \neq j}} x_{k,i,j} M_{ij} + FIXED_TIME_TRUCK \quad (3)$$

$$T_{drone} = \frac{60}{DRONE_SPEED} \sum_{d \in U} \sum_{\substack{i,j \in N \\ i \neq j}} y_{d,i,j} D_{ij} + FIXED_TIME_DRONE \quad (4)$$

$$C_{truck} = \sum_{k \in V} \sum_{\substack{i,j \in N \\ i \neq j}} x_{k,i,j} (M_{ij} \cdot TRUCK_COST) + |V| \times FIXED_COST_TRUCK \quad (5)$$

$$C_{drone} = \sum_{d \in U} \sum_{\substack{i,j \in N \\ i \neq j}} y_{d,i,j} (D_{ij} \cdot DRONE_COST) + |U| \times FIXED_COST_DRONE \quad (6)$$

$$\sum_{k \in V} \sum_{\substack{i \in N \\ i \neq j}} x_{k,i,j} + \sum_{d \in U} \sum_{\substack{i \in N \\ i \neq j}} y_{d,i,j} = 1, \forall j \in N \setminus \{0\} \quad (7)$$

$$\sum_{\substack{j \in N \\ j \neq 0}} x_{k,0,j} = 1, \sum_{\substack{i \in N \\ i \neq 0}} x_{k,i,0} = 1 \quad (8)$$

$$\sum_{\substack{j \in N \\ j \neq i}} x_{k,i,j} - \sum_{\substack{j \in N \\ j \neq i}} x_{k,j,i} = 0 \quad (9)$$

$$\sum_{\substack{j \in N \\ j \neq 0}} y_{d,0,j} = 1, \sum_{\substack{i \in N \\ i \neq 0}} y_{d,i,0} = 1 \quad (10)$$

$$\sum_{\substack{j \in N \\ j \neq i}} y_{d,i,j} - \sum_{\substack{j \in N \\ j \neq i}} y_{d,j,i} = 0 \quad (11)$$

$$u_{k,i} - u_{k,j} + n\, x_{k,i,j} \leq n - 1 \quad (12)$$

$$v_{d,i} - v_{d,j} + n\, y_{d,i,j} \leq n - 1 \quad (13)$$

$$\sum_{i \in N \setminus \{0\}} q_i \left(\sum_{\substack{j \in N \\ j \neq i}} x_{k,i,j} \right) \leq TRUCK_CAPACITY \tag{14}$$

$$\sum_{i \in N \setminus \{0\}} q_i \left(\sum_{\substack{j \in N \\ j \neq i}} y_{d,i,j} \right) \leq DRONE_CAPACITY \tag{15}$$

$$\sum_{\substack{i,j \in N \\ i \neq j}} D_{ij}\, y_{d,i,j} \leq DRONE_RANGE \tag{16}$$

$$x_{k,i,j}, y_{d,i,j} \in \{0,1\}, u_{k,i} \geq 0, v_{d,i} \geq 0 \tag{17}$$

Equation (1) is the total objective function, which integrates the weighted sum of distribution time and transportation cost to minimize the comprehensive efficiency index of distribution. Formula (2) for the distribution of the total time calculation formula, collaborative distribution group to take the ground vehicle distribution time and the larger value of the drone distribution time, the other two groups directly take the corresponding distribution time of a single group. Equation (3) is the specific formula for the distribution time of ground vehicles, the ground vehicles calculate the traveling path with Manhattan distance, and add the fixed distribution time to reflect the actual distribution operation. Equation (4) is the specific formula for drone distribution time, the drone uses the Euclidean distance straight-line flight path, and also adds a fixed distribution time. Equation (5) is the formula for the ground vehicle distribution cost, including Manhattan distance path cost and fixed cost, where the unit distance cost and the number of vehicles together determine the total cost. Equation (6) is the formula for drone delivery cost, including Euclidean distance flight path cost and fixed cost, where drone sorties determine the total fixed cost. Equation. (7) is the customer delivery uniqueness constraint, which ensures that each customer point is visited only once by one delivery mode (ground vehicle or UAV). Equation (8) and (9) represent the departure and return constraints and the flow balance constraints for ground vehicles, respectively, which ensure that the vehicles start at the distribution

center and eventually return, and also ensure that their routes at intermediate nodes are continuous and uninterrupted. Equation (10) and (11) represent the departure and return constraints and flow balance constraints for the UAVs, ensuring the integrity of the UAV routes, which always start at the distribution center and return, while the paths are not interrupted. Equation (12) and (13) are MTZ sub-circuit elimination constraints, which restrict the path structure of ground vehicles and UAVs by introducing auxiliary variables $u_{k,i}$ and $v_{d,i}$ to prevent isolated sub-circuits from being created in the path. Equation (14) and (15) represent the capacity constraints of ground vehicles and UAVs, respectively, which limit the total demand of distribution they undertake to not exceed the maximum load capacity. Eq. (16) is the UAV range constraint, which restricts the sum of the distances flown by the UAVs to not exceed the maximum range limit. Equation (17) is the variable value constraint, which specifies the value ranges of binary decision variables $x_{k,i,j}$, $y_{d,i,j}$ and auxiliary continuous variables $u_{k,i}$ and $v_{d,i}$ to ensure the mathematical completeness of the model and the validity of the solution.

In the model construction of the three groups, the pure vehicle group model uses Equation (1), (2), (3), (5), (7), (8), (9), (12), (14), and (17); the pure UAV group model uses Equation (1), (2), (4), (6), (7), (10), (11), (13), (15), (16), and (17); and the collaborative distribution group adopts all the formulas and constraints from Eq. (1) to Eq. (17) in a complete way, which comprehensively reflects the actual demand and constraint logic of parallel collaborative distribution of UAVs and ground vehicles.

4. Algorithm Design

4.1 Overall design of the algorithm

Since the joint distribution optimization problem involved in this study is an NP-

hard problem, its computational complexity increases exponentially with the increase of the number of nodes, and it is difficult for the exact algorithm to obtain the optimal solution in a short time. Therefore, in order to successfully increase the efficiency and accuracy of the solution, this study uses a multi-stage, step-by-step optimization technique that blends exact and heuristic algorithms. Specifically, firstly, a mixed integer linear programming (MILP) model framework is constructed for the pure ground vehicle group, the pure UAV group and the UAV-ground vehicle collaborative distribution group to unify the evaluation criteria of the distribution efficiency, and the optimal baseline solutions are obtained for the pure ground vehicle group and the pure UAV group with the help of the Gurobi solver; Secondly, in the pure ground vehicle group and the collaborative distribution group, the nearest neighbor heuristic algorithm of the Traveling Provider Problem (TSP) is used to construct the initial feasible solutions quickly, so as to improve the efficiency of population initialization. Subsequently, for the complex collaborative distribution problem, the global path optimization is carried out by the Genetic Algorithm (GA), and the path scheme is continuously improved by selection, crossover, mutation and local search during the algorithm iteration process. In addition, the Warm Start initialization strategy is used to introduce high-quality solutions into the initial population of the genetic algorithm to enhance the solving effect and convergence performance of the algorithm at the initial stage. Finally, during the process of the genetic algorithm and the evaluation stage of the scheme, the Gurobi solver is utilized to accurately solve the embedded complex constraint subproblems and verify the scheme, to ensure that the actual feasibility of the distribution path scheme and the fulfillment of the constraints are achieved, and to guarantee the practical operation and practical value of the proposed scheme. It ensures the practical feasibility of the distribution path scheme and the fulfillment of constraints, and guarantees that the proposed scheme is practicable and practical.

The overall structure of the algorithm is shown in Fig. 4.

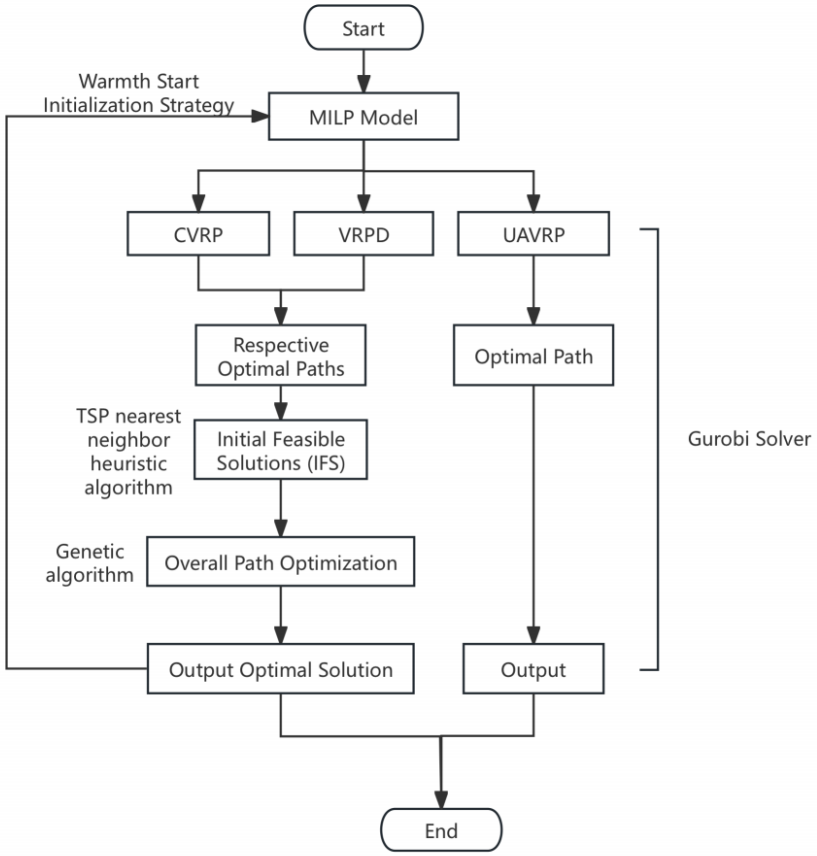


Fig. 4. Algorithm structure block diagram

4.2 Mixed Integer Linear Programming (MILP)

MILP is a mathematical modeling technique for describing optimization problems with linear objective functions and linear constraints, where the variables contain both continuous and integer variables. The specific steps are as follows:

Step 1: For the ground vehicle CVRP mode, UAV UAVRP mode and VRPD mode, the distribution decision variables are first defined based on the MILP method.

Step 2: Set the optimization objective by linear function form to minimize the total cost and total time of distribution. Specifically, multiply and sum each path distance or traveling time with the decision variables to solve the total traveling

distance or total time consumed in the vehicle + UAV delivery mode.

Step 3: Introduce a series of linear constraints using the MILP framework to ensure the feasibility of the solution and satisfy the actual business rules.

Step 4: Model solving and result acquisition. After the variables, objective function and constraints are set, the Gurobi solver is invoked to solve the MILP model optimally.

4.3 Gurobi solver applications

The Gurobi solver is an industry-leading mixed-integer linear programming solver. It uses branch-and-bound, branch-and-cut plane, and multiple heuristics to efficiently solve MILP models. The steps are as follows:

Step 1: First, create an optimization model object through Gurobi's Python interface and add decision variables and parameters. Input the variables and constraints of the aforementioned MILP model into the Gurobi solver, including the path selection variable, linear constraints such as capacity and flight, and the distribution cost objective function.

Step 2: Before running the Gurobi optimization, customize the solution by setting the relevant parameters according to the problem size and solution requirements. This study configures parameters such as Time Limit, Optimality Gap Threshold (MIPGap), Heuristic Strength, and enable/disable Presolve or the number of parallel threads.

Step 3: Using its core algorithm, Branch-and-Bound: The solver first solves the linear relaxation problem ignoring integer constraints to obtain a lower bound, then chooses a branching variable in the solution space to split the problem into

two subproblems (branches), adding constraints that the variable takes either 0 or 1, respectively. Gurobi continues to solve the linear relaxation for each subproblem to obtain a lower bound, and performs a comparative pruning (Bounding) with a global upper bound: if the optimal lower bound for a subproblem is not lower than the currently known optimal upper bound, the subtree does not need to be searched further (node pruning). This iterative process, traversing some of the key branches to determine the optimal solution.

Step 4: During the branch delimitation process, Gurobi dynamically adds Cutting Planes to strengthen the LP relaxation model. At the same time, Gurobi uses various pre-solution techniques to simplify the model, such as fixing variables, removing redundant constraints, and simplifying matrix coefficients, so as to reduce the number of branching nodes and improve the efficiency of the solution.

Step 5: Gurobi combines heuristic algorithms to find feasible solutions as upper bounds during the search process. During MILP optimization, Gurobi’s built-in heuristics generate feasible integer solutions from the current LP relaxation, updating the incumbent solution when an improved objective is found.. If a new feasible solution is found and its objective value is better than the current upper bound, the global upper bound (optimal solution candidate) is updated.

Step 6: The optimal solution is generated when Gurobi stops the search by satisfying the optimality condition or by reaching a preset stopping criterion (e.g., time exhaustion).

4.4 Traveler's Nearest Neighbor Heuristic Algorithm (TSP)

The nearest neighbor heuristic algorithm is a greedy strategy for constructing the initial distribution path. The algorithm starts from the starting point and selects the unvisited customer with the shortest distance from the current node as the next

access point at each step, and finally forms a closed loop. The specific steps are as follows:

Step 1: First, the distribution center (the starting node) is selected as the current node, and an empty path and a set of unvisited customers are initialized. The algorithm initializes the route at the distribution center and iteratively expands the path by selecting the nearest unvisited customer.

Step 2: In each iteration, from the current node, find the next customer node in the set of unvisited customers that is closest to it. Using the pre-computed distance matrix or coordinate distances from the local greedy strategy, a minimum distance filter is performed in the unvisited list to select the nearest customer point to the current node.

Step 3: Add the selected nearest neighbor customer node to the current path sequence and mark the customer as visited and remove it from the unvisited set. Then, update the current node to the customer node that has just been added so that the route can continue to be constructed from that node in the next step.

Step 4: Closing the route. Repeat Steps 2 and 3 to iteratively select the next closest customer and extend the path until all customer nodes are marked as visited. Finally, add the distribution center back to the end of the path to close the loop and return the route to its starting point to complete the distribution. Finally, a complete route generated with the nearest neighbor rule is returned.

The flow chart of Traveler's nearest neighbor heuristic algorithm is shown in Fig. 5.

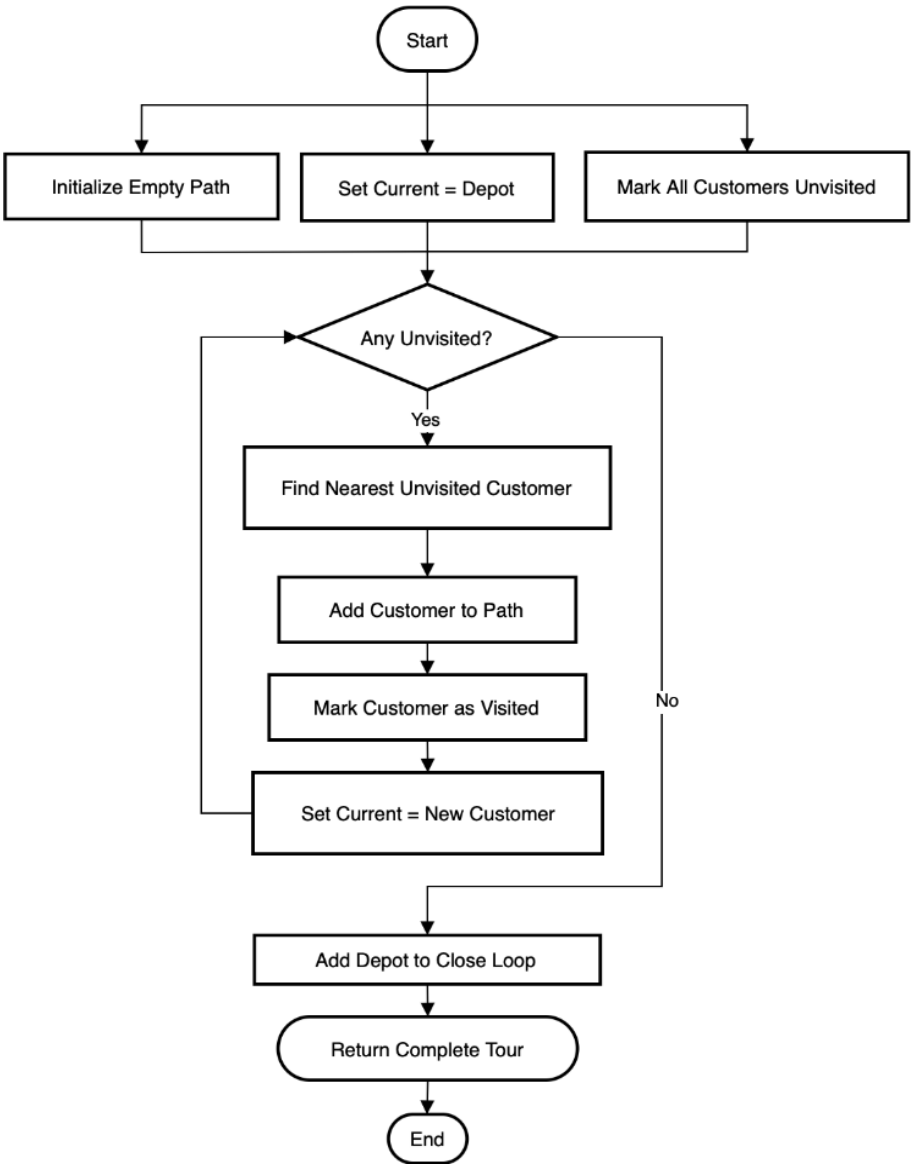


Fig. 5. Flow chart of nearest neighbor heuristic algorithm

4.5 Genetic Algorithm

Genetic algorithm is a global stochastic search method based on natural evolutionary principles for solving combinatorial optimization problems. The specific steps are as follows:

Step 1: Generate an initial population of solutions, where each individual encodes a complete assignment of customer visits for all vehicles and drones based on the problem dimensions.

Step 2: For each solution individual in the initial population, call the function to check and fix whether it violates the constraints. Since randomly generated or cross-generated solutions may appear to not satisfy constraints such as capacity, range, etc., the specific module performs a constraint correction operation for each individual.

Step 3: Calculate the fitness value for each repaired individual to measure the merit of its solution. The fitness function is consistent with the optimization objective, and the negative value of the total cost or time of distribution is usually used as the fitness (the lower the cost, the higher the fitness)

Step 4: A tournament selection strategy is used for optimal replication of the population using the fitness value. A number of individuals from the current population are randomly selected for the "tournament", their fitness is compared, and the individual with the highest fitness is selected to enter the parent pool. The process is repeated until the number of parents required for the new generation has been selected. The algorithm ensures that the best solutions with higher fitness (and lower cost) have a higher probability of being selected as parents to pass on their features through a specific function module, keeping the evolutionary direction converging towards the better solutions.

Step 5: For generating new offspring solutions, genetic algorithm applies the crossover operator to the selected parent-individual pairs. The code implements the crossover operation by means of a specific function that randomly selects the coded segments of the distribution routes of the two parents and exchanges these two segments, thus reorganizing the path structure of the parent solution in the offspring.

Step 6: To maintain population diversity and avoid premature convergence, the algorithm applies randomized mutations to the offspring solutions generated by

crossover. Through the offspring variation, the individual distribution solutions are slightly modified to prevent the genetic algorithm from falling into a local optimum.

Step 7: For the new solution individual obtained after crossover and mutation, the local search function is further invoked to perform local optimization and improve the quality of the solution. Traverse the local neighborhood of the offspring solutions and replace the current solution if a better neighborhood solution is found.

Step 8: After completing the local search, the resulting new offspring solutions are merged into the population to form the next generation solution set. The intergenerational replacement strategy is adopted: the optimal individuals in the previous generation are retained according to the fitness (elite retention), and the inferior individuals are replaced with the new subgeneration to ensure that the size of the population is stable while gradually improving the average fitness. Then the algorithm determines the termination condition, judging whether the current number of generations reaches the preset maximum number of iterations, or the optimal fitness gain of the population is less than a threshold value (convergence criterion) within a number of generations. If the termination conditions are not met, the new population is used as the current population, and return to step 4 to continue the execution of the selection, crossover, mutation, local search cycle iteration. In this way, the genetic algorithm evolves generation by generation, constantly approaching the globally optimal collaborative distribution route scheme. When the termination conditions are satisfied, the algorithm outputs the individual with the highest fitness in the population as the final near-optimal solution, realizing the global optimization solution of the collaborative distribution mode.

The flow chart of genetic algorithm is shown in Fig. 6.

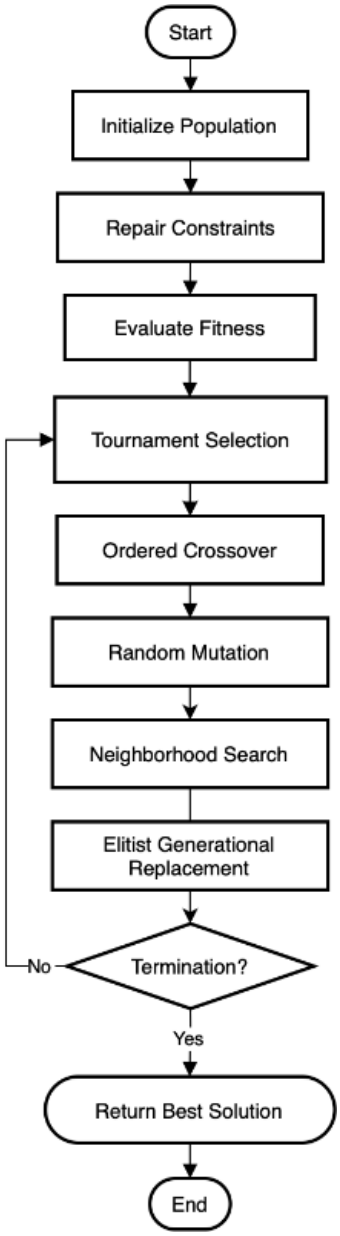


Fig. 6. Flow chart of genetic algorithm

4.6 Warm Start

The Warm Start initialization strategy is a method to speed up the solution process by introducing high quality initial solutions. The specific steps are as follows:

Step 1: Combine the pure vehicle routes obtained through MILP solving with the pure UAV routes to construct an initial collaborative distribution scheme, and

directly adopt the feasible routes obtained by the nearest neighbor algorithm as candidate solutions.

Step 2: The selected a priori high-quality solution is added to the initial population of the genetic algorithm as a seed individual. In the initial population phase of the realization, the solution is directly encoded and inserted into the population list, thus partially replacing the originally purely randomly generated individuals.

Step 3: In contrast to the entirely random initialization, the used high-quality genetic solution is optimized to provide a higher level of initial fitness. This enables the crossover and selection processes to combine high-quality offspring more quickly and speeds up the approximation of the global optimum.

5. Case study in Shenzhen

5.1 Parameter settings

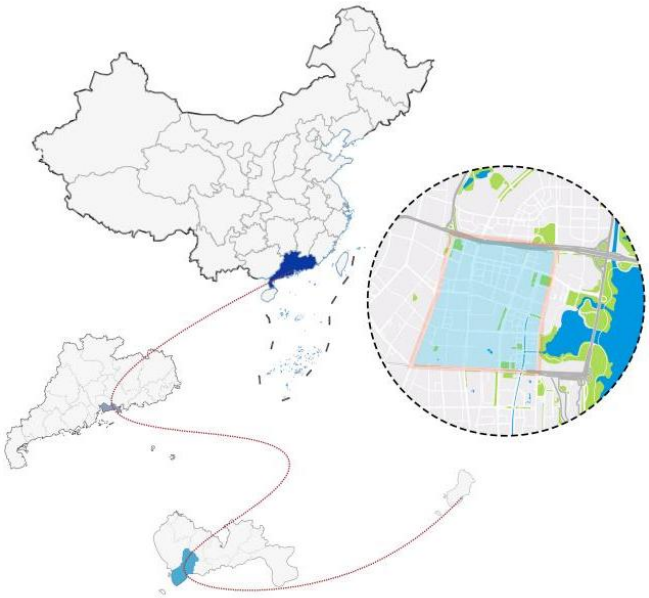


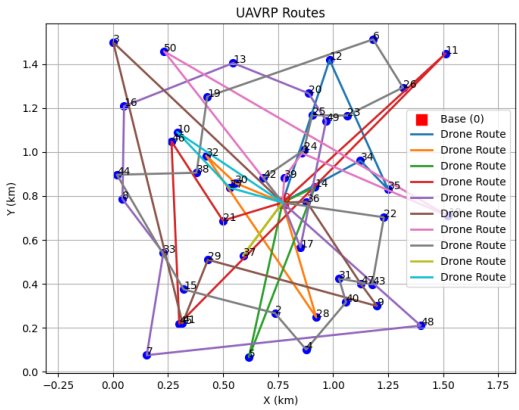
Fig. 7. Geographic Location of the Case Study Area

This practical case is solved by python throughout. Due to the representative mixed-use urban fabric of this area and its status as a pilot region for Meituan's drone delivery and autonomous vehicle delivery services, etc, about 2.37 km^2 of the area around Shenzhen Talent Park is selected as the experimental range, as shown in Fig. 7. Simulate its regional scope in the experimental tool, generating a rectangular area with a length of 1.75 km and a width of 1.4 km. Based on the fact that the selected area is mostly commercial complexes and office buildings, the same 50 random customer points are generated in three groups located in the experimental scope, and the simulation distribution center is located in the center of the area. The parameter settings of this case study are in line with the model theory and the current situation of the industry while combining with the reality, a series of settings and corrections have been made. The Euclidean distance is used for UAV flight distance and Manhattan distance is used for vehicle traveling distance. Each customer's demand is capped at 4kg, assuming good road conditions, a maximum vehicle load of 60kg, a maximum UAV load of 20kg, a maximum UAV flight distance of 5km, an average flight speed of 60km/hour, and an average vehicle travel speed of 20km/hour. The average traveling speed of the vehicle is 20 kilometers per hour. The fixed delivery time of the bus is 5 minutes, and the fixed delivery time of the drone is 4 minutes. The unit cost of the drone is 250,000 CNY and the unit cost of the ground vehicle is 500,000 CNY. According to the Civil Aviation Management Institute of China (2024), the model sets drone unit distance cost per kilometer 10,000 CNY, ground vehicle unit distance cost per kilometer 8000 CNY. In reference to The economics of urban transportation(2007), value of time is typically proxied by the average wage rate, representing the amount of money individuals are willing to exchange for time savings. Meantime, considering that urban delivery involves not only operators but also multiple stakeholders—such as platform managers, maintenance personnel, and end users—the time cost conversion factor is adjusted to reflect the composite labor value across all parties involved, the time conversion factor calculated as 1 thousand CNY per 1 minute. Considering that time is usually

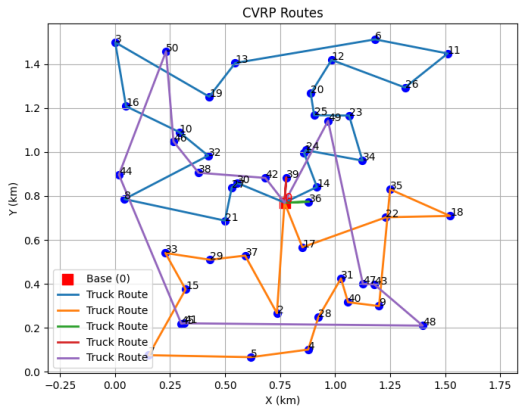
weighted higher than monetary costs in end-of-line delivery and that consumers are highly sensitive to delivery timeliness, the time weight is 0.8 and the cost weight is 0.2.

5.2 Results

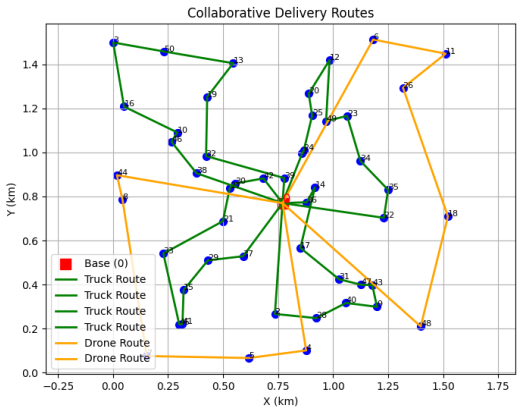
The final solution times of the three groups are 3.89 seconds, 3.24 seconds, and 1800 seconds, respectively, with optimality gap thresholds of 1.59%-1.87%, which does not trigger the penalty mechanism, and the solution quality is high. The visualized distribution results in different scenarios are shown in Fig. 8. It can be seen that the pure ground vehicle mode has the highest total distance and time due to the circuitous path; In terms of route composition, the VRPD mode subdivided the distribution paths of ground vehicles into four, and the distribution paths of drones into only two, covering a number of nodes with scattered locations but relatively small demand; whereas in the CVRP mode, the vehicles have to undertake the delivery of all the nodes, forming up to five longer paths, and a considerable part of the nodes need to be detoured to the far end and then return to the base, which increases the overall mileage and route and cost. As for the UAVRP mode, the UAV has the largest number of paths and needs to make multiple round trips to complete the delivery of all the nodes, and some of the nodes are split into multiple segments due to load and range limitations, which significantly increases the flight distance. It can be seen that VRPD not only significantly reduces the ineffective driving of vehicles on remote routes, but also reduces the cumulative flights of drones with high intensity and multiple batches by assigning the core areas with high demand to ground vehicles for centralized handling, and letting drones take care of the scattered or remote segments exclusively.



(1) UAVRP results



(2) CVRP results



(3) VRPD results

Fig. 8. Visualized distribution results in different scenarios

Comparison of the results is shown in the Table 1. The pure UAV mode has a direct path but is limited by the load and endurance, and its multiple flights increase the total cost; and the collaborative distribution mode gives full play to the advantages of the vehicle's large capacity and the flexibility of the UAV's delivery, and realizes a significant reduction of the total path length, distribution time, and transportation cost. The total path length, delivery time and transportation cost are significantly reduced.

Since the objective function of this experiment is the minimum value obtained from the cost and time weighting calculation, the smaller the efficiency parameter is, the higher the actual efficiency is. The experimental results show that the

vehicle-drone collaborative mode exhibits optimal performance in all indicators. In terms of distance traveled, the total distance traveled by the vehicle is reduced by 16% compared to the vehicle-only mode and 43% compared to the drone-only mode. In terms of traveling time, the total delivery completion time is 42% lower than that of the pure ground vehicle group, and 10% higher than that of the pure drone group, because the vehicles and drones are able to perform their respective delivery tasks at the same time. Compared to a ground-only vehicle, the overall cost of delivery is 2% lower, and compared to a drone-only vehicle, it is 50% lower. While the expense of using drones is efficiently employed with planned scheduling, the decrease of vehicle mileage and time saves a significant amount of money on labor and fuel. Every drone flight eliminates pointless round trips and covers more delivery locations. When combined, the synergistic model achieves the highest delivery efficiency achievable, increasing delivery efficiency by 4% compared to group of ground-only vehicles and 5% compared to group of drones alone. This case study verifies the superiority of the collaborative distribution mode in real-world scenarios, confirming the value of the model and algorithm proposed in this paper.

Table 1. Comparison of three groups routes and results

Delivery Mode	Solution Method		Delivery Routes	Delivery Distance /km	Travel Cost
VRPD	Two-stage solution	Truck	0→38→46→10→16→3→50→13→19→32→39→0 0→37→29→15→41→45→33→21→27→30→42→0 0→22→35→34→23→49→12→20→25→24→1→0 0→36→14→31→47→43→9→40→28→4→17→0	16.80	14.68

		Drone	0→2→5→7→8→44→0 0→48→18→26→11→6→0		
CVRP	Two-stage solution	Truck	0→30→27→21→8→32→10→16 →3→19→13→6→11→26→12→2 0→25→23→34→24→1→14→0 0→2→37→29→33→15→7→5→4 →28→31→40→9→35→18→22→ 17→0 0→36→0 0→39→0 0→49→47→43→48→41→45→44 →50→46→38→42→0	19.20	15.36
UAVRP	MILP Model	Drone	0→34→35→12→0 0→30→32→28→0 0→14→5→0 0→21→46→45→11→0	29.48	29.48

Continued Table 1

UAVRP	MILP Model	Drone	0→17→49→20→13→16→8→33 →7→48→0 0→36→47→9→29→41→3→0 0→39→1→18→50→0 0→42→24→25→23→26→6→19 →38→44→15→2→4→40→31→4 3→22→0 0→37→0 0→27→10→0	29.48	29.48

5.3 Sensitivity test analysis

In this test, due to the upper limit of the maximum load at a fixed customer point as well as the load weight and initial cost of the carrier, the model performs a sensitivity test on the key parameters (UAV flight speed, weight allocation of delivery time and cost, and time conversion factor) to analyze the impact of changes in each parameter on the delivery results. As shown in the Table 2, the results show that the model is most sensitive to changes in the target weighting parameters, followed by the UAV speed, and the time conversion factor is the least sensitive. However, the efficiency of the synergistic group is still the highest regardless of the changes. When the time weight is set to 0.8 and the cost weight is set to 0.2, the distribution efficiency of the collaborative distribution mode (VRPD) (55.88) is improved by about 3.79% compared with that of the vehicle-only group (CVRP, 58.08); when adjusting the time conversion coefficient to reduce it from 0.1 to 0.01, the efficiency of the VRPD mode decreases from 55.88 to 53.25, with a efficiency promotion of 4.71%, at which point the efficiency of the VRPD mode is improved by about 0.60% compared to the CVRP mode (53.57). After further increasing the time weight to 0.9 and reducing the cost weight to 0.1, the distribution efficiency advantage of the collaborative distribution mode VRPD is even more significant (30.11 vs. 32.17), with an efficiency improvement of 6.40%. In addition, when the UAV speed is increased from 60 km/h to 80 km/h, the efficiency of coordinated distribution mode (56.05) is improved by about 3.50% compared with the pure vehicle distribution mode (58.08), which indicates that increasing the flight speed of the UAV has a significant effect on the efficiency improvement of coordinated distribution. At the same time, The robustness and reliability of the model under different parameter variations are also demonstrated, showing that it is more tolerant to external environment or data fluctuations. Overall, the addition of UAVs to vehicle distribution is equivalent to the addition of faster and lower-cost distribution tools, which helps to reduce the overall distribution time and cost.

Table 2. Distribution effect of different experimental groups

Delivery Mode	Fixed cost	Travel cost	Total cost	Total time /min	Delivery Efficiency (Weighted)
UAVRP	250	29.48	279.48	33.48	58.57
CVRP	250	15.36	265.36	62.59	58.08
VRPD	250	14.68	264.68	36.78	55.88

Group Number	Delivery Mode	Drone speed/ km/h	Truck speed/ km/h	Travel Cost	Total Cost	Total Time/ min	Time weight	Cost weight	Time conversion factor	Delivery Efficiency
1	UAVRP	60	20	29.48	279.48	33.48	0.8	0.2	0.1	58.57
	CVRP			15.36	265.36	62.59				58.08
	VRPD			14.68	264.68	36.78				55.88
2	UAVRP	60	20	29.48	279.48	33.48	0.8	0.2	0.01	56.16
	CVRP			15.56	265.56	62.59				53.57
	VRPD			14.81	264.81	36.51				53.25
3	UAVRP	60	20	29.48	279.48	33.48	0.8	0.2	0.05	57.23

Continued Table 2

3	CVRP	60	20	15.36	265.36	62.59	0.8	0.2	0.05	55.58
	VRPD			14.68	264.68	36.78				54.41
4	UAVRP	60	20	29.48	279.48	33.48	0.9	0.1	0.1	30.96
	CVRP			15.36	265.36	62.59				32.17
	VRPD			15.83	265.83	39.18				30.11
5	UAVRP	60	20	29.48	279.48	33.48	0.7	0.3	0.1	86.19
	CVRP			15.36	265.36	62.59				83.99
	VRPD			15.50	265.50	39.87				82.44
6	UAVRP	60	20	29.48	279.48	33.48	0.6	0.4	0.1	113.80
	CVRP			15.36	265.36	62.59				109.90
	VRPD			13.63	263.63	43.83				108.08
7	UAVRP	75	20	29.48	279.48	27.59	0.8	0.2	0.1	58.10
	CVRP			15.36	265.36	62.59				58.08
	VRPD			15.21	265.21	38.79				56.15
8	UAVRP	70	20	29.48	279.48	29.27	0.8	0.2	0.1	58.24
	CVRP			15.36	265.36	62.59				58.08
	VRPD			14.89	264.89	40.46				56.22
9	UAVRP	80	20	29.48	279.48	26.11	0.8	0.2	0.1	57.99
	CVRP			15.36	265.36	62.59				58.08
	VRPD			14.93	264.93	38.29				56.05

6. Discussion

Our experimental results are closely related to the research objectives in the literature review. As mentioned earlier, the traditional ground logistics system has a limited ability to solve some special needs, i.e. when there are emergencies and time sensitive issues. In view of this, the low-altitude economy can be targeted to solve these problems to improve a three-dimensional urban logistics system. The experimental results show that the VRPD mode improves the overall efficiency

(reducing the weighted cost-time index by ~4-5%), and reduces total delivery time and cost by about 4% and 5%, respectively. This is consistent with the theoretical framework of a parallel delivery strategy proposed above (Murray & Chu, 2015), which has clarified the feasibility of VRPD application in some practical scenarios.

In the practical application scenario, the results of this experiment and the results of sensitivity test support the application of low altitude economy to the actual logistics system to build the framework of urban intelligent logistics. For example, the place we investigated, Shenzhen, has proposed the "Low Altitude Economy and Industry Promotion Regulations 2024" at the policy level to support the promotion of low-altitude economy-related infrastructure construction and improve the airspace management of UAVs. In addition, in terms of limitations, the mode does not fully consider the impact of urban three-dimensional spatial obstacles, such as some high buildings on UAV paths. It is similar to the simplified hypothesis of Li and Xin (2024). Thus, further research is needed to verify the robustness of the model with real road network data.

In terms of social impacts, the results of our experimental groups show that the low-altitude logistics collaborative distribution mode based on UAV technology has improved some limitations of the traditional logistics distribution mode with high labor cost, low efficiency and limited coverage of a single transportation. Therefore, our research provides quantitative support for the positive social impact of this efficient collaborative distribution mode on the urban logistics field. This shows the huge market potential in China's low-altitude economy industry. In addition, our research is also conducive to deepening the public recognition and further exploration of its application value. Therefore, it is beneficial to promote the development of this collaborative distribution mode to a more flexible, efficient and wider field. More specifically, within the scope of our research, we believe that based on the collaborative distribution mode, the

application of low-altitude logistics technology and equipment will further improve comprehensive efficiency in our experiment, and shorten distribution time. In this way, it can promote the optimization and upgrading of the logistics industry structure, improve the distribution efficiency and the convenience of people's lives and other social impacts.

In policy implications, national and local units, especially in Shenzhen, Hangzhou, Shanghai and other leading cities, have taken a positive attitude towards unmanned distribution, and even introduced preferential subsidy policies in some areas to accelerate its development. What’s more, before unmanned distribution was put into use, that is to say, in the early stage of the development of the distribution mode, the state introduced more macro encouragement and development plans. After unmanned delivery gradually enters the trial stage, our research and experimental results can provide quantitative data and efficiency comparison results support for local governments, especially Shenzhen Talent Park and its surrounding areas. So, these will promote the generalization of collaborative distribution mode and the formulation and implementation of specific management plans, which can promote relevant policies to be more directional and regional.

In terms of urban traffic impacts, the collaborative UAV–ground vehicle distribution mode can alleviate road congestion by shifting a portion of delivery traffic to the air. By reducing the overall distance and number of trips made by ground vehicles, UAVs can handle some last-mile deliveries, potentially reducing bottlenecks and delivery traffic during peak hours. These results subtly support a traffic effect analysis of the new model by showing that incorporating low-altitude deliveries can lessen the strain on urban road networks, even if the main focus of our study is logistics efficiency. This insight not only reinforces the environmental and social benefits discussed above (such as reduced emissions and expanded service coverage), but also aligns with the theoretical notion of a

three-dimensional urban transport network that enhances overall city traffic conditions. Therefore, the low-altitude economy, as evidenced by our results, holds strategic value for urban planning by complementing traditional ground transport and contributing to more resilient urban traffic systems.

7. Recommendation

This study provides empirical and algorithm model support for the standardization and efficiency of low-altitude economic development. The case analysis results show that VRPD mode can significantly improve logistics efficiency, and the specific data is shown above. Therefore, these data all point to the policy suggestions on promoting the reform of UAV airspace management and innovative collaborative distribution mode, and provide a quantitative basis. Shenzhen took the lead in establishing a hierarchical and classified management airspace system and subsidizing enterprises that innovate collaborative distribution modes to guide the direction. In addition, according to the literature review, in the process of collaborative delivery, it is necessary to improve the construction of low-altitude flight service support and improve the UAV traffic management system, so as to enhance the reliability of collaborative delivery and improve the distribution efficiency. Furthermore, it is also consistent with the goal of UAM program (FAA, 2023) of the Federal Aviation Administration (FAA) mentioned above.

In terms of algorithm framework, although the overall algorithm flow of this study shows good solving efficiency, there are still some limitations that need to be improved. In terms of algorithm performance, TSP nearest neighbor algorithm and genetic algorithm have high computational efficiency, but they are easy to fall into the problem of local optimal solution. Therefore, to solve this problem, algorithms such as simulated annealing can be considered. Moreover, in the case simulation, the modeling of the real 3D path network is still relatively simplified.

Especially, it will be influenced by the urban canyon effect and road resistance coefficient. To solve this problem, it is recommended to use the Solomon benchmark data set and the real road network data to improve the verification process, especially the three-dimensional building model and real-time meteorological data. Furtherly, it can be used to improve the environment adaptability of the path planning algorithm.

In addition, the model does not fully consider the differences in customer demand, adopts a unified distribution strategy for all customers, which may affect the efficiency of mixed distribution of light and heavy goods. So, the weight ratio of priority points can be added before setting the algorithm. Overall, these factors will affect the practical application effect, so, it is necessary to combine the actual data of low-altitude economy for more accurate algorithms and realistic consideration.

In the application of various industry segments, the collaborative delivery mode shows higher efficiency and flexibility. Specifically, in realistic scenarios with high timeliness requirements, such as medical emergency supplies distribution, fresh cold chain and other fields, VRPD mode needs to be prioritized. More importantly, its algorithm parameters can be flexibly adjusted according to the realistic needs in these scenarios, thereby improving applicability and distribution efficiency.

In the specific cooperation mode, the ground vehicle can be mainly responsible for the distribution of high-density areas, and UAV can focus on solving the decentralized distribution problem at the end of the real path. Of course, it can also be cross-applied according to the actual application scenario. Specifically, in order to achieve the large-scale landing of specific application scenarios, it is necessary to build a new infrastructure network including distributed storage nodes, drone charging stations, airspace communication base stations, etc., to

smooth the entire distribution path. This requires that the government, enterprises and technology suppliers enhance the cooperation, and jointly promote the development of standards, resource integration and business model innovation.

8. Conclusions

This research builds an algorithm optimization model for collaborative delivery between UAVs and ground vehicles. The model combines mixed integer linear programming (MILP) and hybrid algorithm, and shows the high efficiency of VRPD in a real case in Shenzhen. Therefore, this not only confirms the problem of insufficient flexibility of the traditional logistics system proposed in the introduction, but also provides algorithm support and practical application methods for low-altitude economy in the field of intelligent logistics.

The innovation of the research is reflected that the hybrid of Warm Start strategy and TSP nearest neighbor heuristic algorithm effectively improves the initial population quality of the genetic algorithm, thus accelerating the convergence process. At the same time, the model uses the fine processing of Manhattan distance and Euclidean distance respectively to calculate the differential distance of UAV and vehicle, which enhances the rationality of path planning. Moreover, these methods provide a useful operational framework for solving NP-hard problems.

In general, through the innovation of algorithm framework and case test, this study fully demonstrates the complementary advantages of UAV and vehicle in the delivery process. In this way, this mode can improve the delivery efficiency of intelligent logistics system and reduce the comprehensive cost. Therefore, future research should focus on optimizing the distribution environment modeling algorithm, fully consider the impact of different environments on the collaborative delivery efficiency of drones and vehicles. More significantly,

future researches are expected to improve the system's adaptability in complex scenarios such as three-dimensional space and dynamic road conditions, and further promote the deep integration of low-altitude economy and urban logistics system, at the same time, deepening its practical application level.

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Thank God the Earth Hasn't Exploded Yet.

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